STUDY FRAMEWORK

Conditional bilateral correspondence: from images to evidence

Test cross-eye correspondence against grade and quality controls; audit view effects separately.

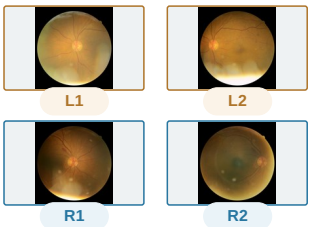
A Patient as the unit

DeepDRiD · 300 complete bilateral patients

SOURCE COHORT

Patient i

2 eyes × 2 views = 4 images



Patient-disjoint train / val / test: 210 / 46 / 44

B Define the contrasts

One source patient; preserve eye and view labels

POSITIVE

L(i) ↔ R(i) same patient, opposite eyes

NEGATIVE

L(i) ↔ R(j) different patients; j ≠ i

View checks

L1–R1 · L1–R2 · L2–R1 · L2–R2

Same-eye repeats (L1–L2, R1–R2) audit hierarchy.

Five fixed patient-level negative assignments

C Measure correspondence

Frozen raw representation · no feature tuning

ImageNet ResNet-50
same fixed encoder for each image

↓

Cosine score s(L, R)

↓

Pair AUROC + cosine gap

Auxiliary: grade / quality metadata probes

D Make cross-patient negatives progressively harder

The positive pairs and frozen features stay fixed; only negative-patient matching changes.

01 DISEASE CONTROL

Same DR grade

Mean |Δ quality| 0.198

grade stratum · patient derangement

→

02 JOINT CONTROL

Grade + exact quality tier

Mean |Δ quality| 0.071

exact grade × tier bucket

→

03 SENSITIVITY

Grade + nearest quality

Mean |Δ quality| 0.024

within grade · nearest assignment

E Audit at three evidence levels

DeepDRiD
INTERNAL MEASUREMENT

Full 300: condition controls
Test: four-view n=43; probes n=44

CFP–OCT
DEVELOPMENT STRESS

137 eligible: raw vs H-CBFD
Candidate loses to raw baseline

PAPILA
LOCKED EXTERNAL BASELINE

244 patients · one image/eye
Frozen raw features only



Claim boundary: correspondence does not establish patient identity or a disease-independent biomarker.